

Part 2 – Time Series Analysis Report

(a) Visual Inspection:

If the series shows persistence, trends, or slow decay patterns, this suggests autocorrelation between weeks.

(c) Model Suggestion:

If ACF tails off gradually and PACF cuts off after lag $p \rightarrow \text{AR}(p)$

If PACF tails off and ACF cuts off after lag $q \rightarrow \text{MA}(q)$

If both tail off $\rightarrow \text{ARMA}(p,q)$

Selected Model: ARIMA(1, 0, 1)

(d) Model Parameters:

Estimate Lower CI Upper CI P-value

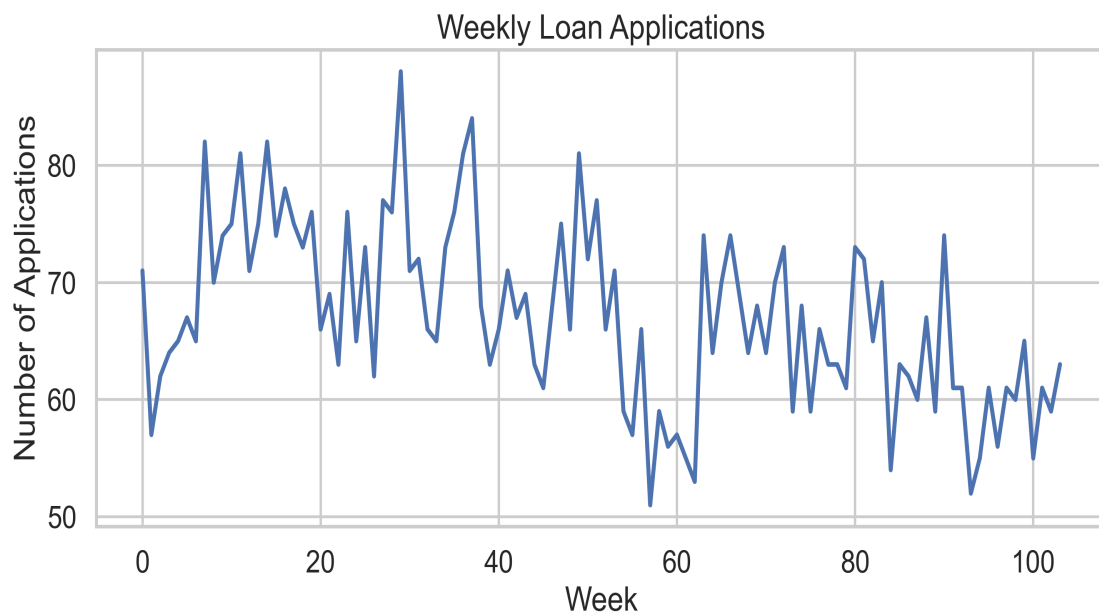
const 66.746024 62.656597 70.835452 1.513606e-224

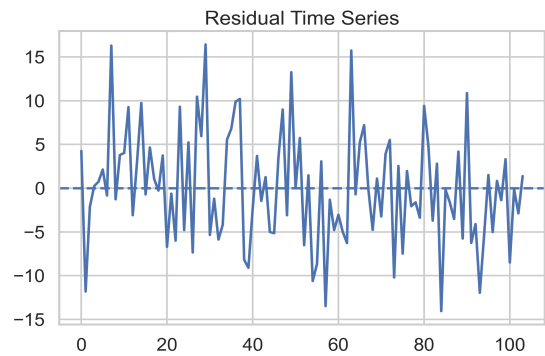
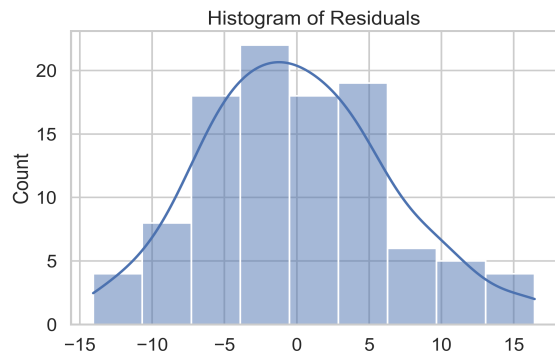
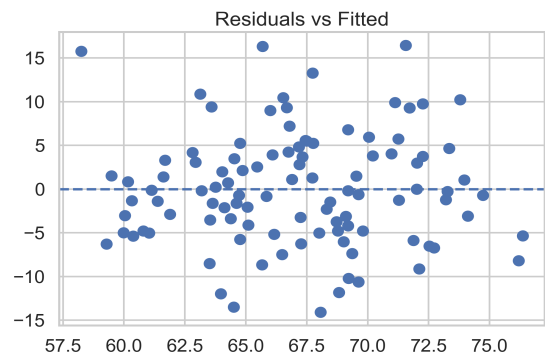
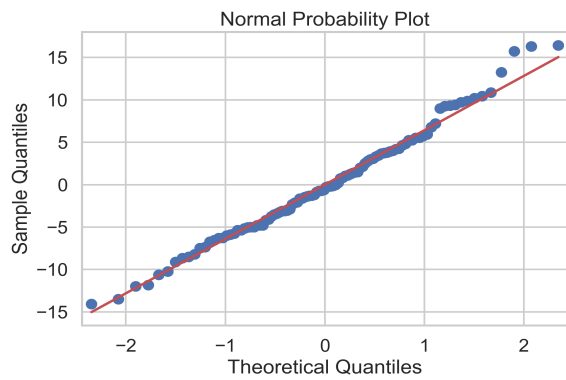
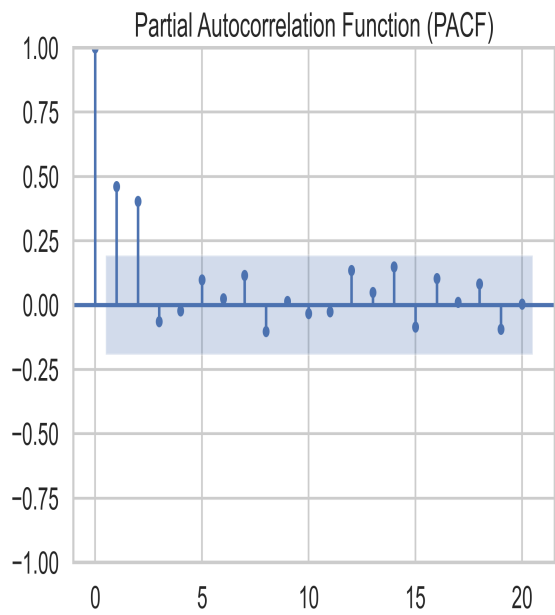
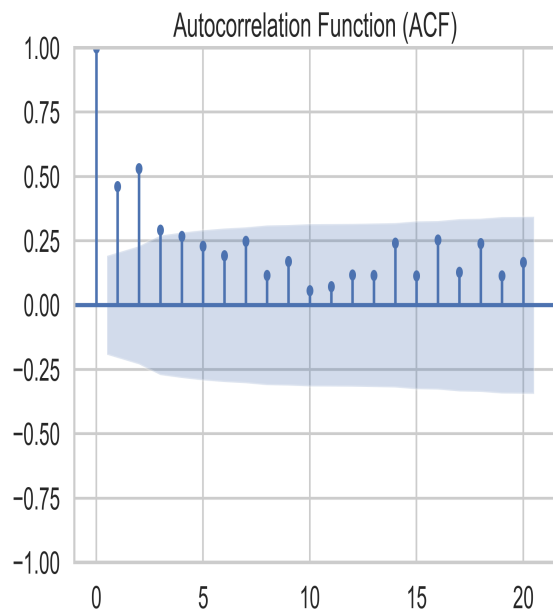
ar.L1 0.857493 0.656058 1.058928 7.218622e-17

ma.L1 -0.524080 -0.874088 -0.174071 3.338460e-03

sigma2 40.970744 29.235352 52.706136 7.774126e-12

(e) Mean Squared Error (MSE): 41.1243





ARIMA Model Parameters:

Parameter	Estimate	Lower CI	Upper CI	P-value
const	66.7460	62.6566	70.8355	0.0000
ar.L1	0.8575	0.6561	1.0589	0.0000

ma.L1	-0.5241	-0.8741	-0.1741	0.0033
sigma2	40.9707	29.2354	52.7061	0.0000

Mean Squared Error (MSE): 41.1243

Sample of Actual vs Fitted Values:

Actual	Fitted
71.0	66.75
57.0	68.82
62.0	64.12
64.0	63.76
65.0	64.27
67.0	64.87
65.0	65.85
82.0	65.69
70.0	71.28
74.0	70.21